Science and Society

Science

May, 2026

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Science and Society (A11411)

Short Title:	Science and Society		
Faculty	Science and Computing		
Department	Science		
Cluster	Environmental Science		
Author	EOWENS		
Credits	5	**Level:**	Introductory

Description of Module / Aims

Science is important to all areas of human life. Understanding science, therefore, is essential to our ability to make informed decisions. This module aims to help students to develop and improve their scientific thinking and skills. Topics such as the development of science in the course of history, the structure of scientific theories and and the interrelationship between science and social, cultural, political and economic structures will be discussed.

Programmes

SCIE-0043	BSc (Common Entry) (WD_SSCIE_D)	1 / 2 / E
SCIE-0043	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 2 / E
SCIE-0043	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 2 / E
SCIE-0043	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 2 / E
SCIE-0043	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 2 / E
SCIE-0043	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 2 / E
SCIE-0043	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 2 / E

Indicative Content

- Introduction to the philosophy of science: what is science?
- Emergence and professionalism of modern science: rise of science and reasons for the rise of science
- Boundaries of science: what is science and what is not?
- Construction of scientific knowledge: essentialist and constructivist approaches
- Introduction to ethics in science
- Review of issues in science and society, e.g. genetically modified (GM) foods, stem cell research, environment versus industry, climate change and the energy crisis etc.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe important episodes in the history of science.
2. Describe how scientific knowledge is acquired and validated.
3. Discuss different modern philosophical approaches to science (Popper, Khun).
4. Discuss different ethical philosophies (Utilitarianism, Deontology, Consequentialism).
5. Apply different ethical approaches to scientific and social debates, for example GM foods, stem cell research, environmental versus industry, climate change, energy crisis.
6. Identify scientific arguments as they appear in scientific and popular texts.

Learning and Teaching Methods

- Lectures.
- Tutorials.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Essay	30%	1,2
Participation	20%	3,4,6
Group Project	50%	5

Assessment Criteria

<40%: Inability to demonstrate a basic understanding of the fundamental principles of the module content.

40%-49%: Ability to demonstrate a basic understanding of the fundamental concepts of the relationship between science and society.

50%-59%: Ability to clearly describe and describe concepts encountered in the module.

60%-69%: Demonstrates a very good understanding of the majority of the module content and providing good reasoning and analysis skills.

70%-100%: All the above to an excellent level and demonstrate excellent reasoning skills and focused well thought out arguments.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Tutorial	24	
Independent Learning	99	

Essential Material

- "Module Moodle page." Science and Society. http://www.moodle.ie

Supplementary Material

- Ede, A. and L.B. Cormack. _A History of Science in Society: From Philosophy to Utility_. 2nd ed.. Canada: University of Toronto Press, 2012.
- Talbot, M. _Bioethics: An Introduction_. 1st ed. USA: Cambridge University Press NY, 2012.